

Bonus/ Extra Credit (20 Pts)

Flask Code GitHub Link: <https://github.com/Rokeya-Siddiqua/COSC502-Final-Project-Spring26/tree/master>

This PDF contains screenshots of the working web application showing search results for different queries.

Example Queries

- angiographic image shows normal coronary artery
- CT scan demonstrating pulmonary embolism
- MRI scan showing brain tumor
- xray image of fractured bone

Medical Image Caption Retrieval System

Medical Image Caption Retrieval System

Top 5 Results for: "MRI scan showing brain tumor"

Result 1

Image ID: ImageCLEFmedCaption_2022_train_000131

Caption: mri of the brain with contrast increase flow in the left cavernous sinus and engorgement of the left ophthalmic vein

Similarity Score: 0.4021

Result 2

Image ID: ImageCLEFmedCaption_2022_train_000840

Caption: computed tomography scan show the presence of a right heart tumor develop on both side of the tricuspid valve

Similarity Score: 0.2907

Result 3

Image ID: ImageCLEFmedCaption_2022_train_000892

Caption: dripinfusion cholangiographycompute tomography scan fortyeight day postoperatively the right accessory hepatic duct be intact black arrow and it dominate an anterior segment of the right hepatic lobe as indicate by the intraoperative cholangiography scan

Similarity Score: 0.2407

Result 4

Image ID: ImageCLEFmedCaption_2022_train_000661

Caption: leave anterior inferior cerebellar artery hyperplasia and cochlear artery hyperplasia selective angiography mri

Similarity Score: 0.1776

Result 5

Image ID: ImageCLEFmedCaption_2022_train_000915

Caption: angiography show ascend pharyngeal artery as tumor feed vessel

Similarity Score: 0.1555

Medical Image Caption Retrieval System

Top 5 Results for: "xray image of fractured bone"

Result 1

Image ID: ImageCLEFmedCaption_2022_train_000992

Caption: percutaneous injection of bone cement combine with particle implantation use a dsa machine guide

Similarity Score: 0.2959

Result 2

Image ID: ImageCLEFmedCaption_2022_train_000986

Caption: xray image of the perfuse porcine kidney

Similarity Score: 0.2543

Result 3

Image ID: ImageCLEFmedCaption_2022_train_000958

Caption: wire place over the par interarticularis on lateral view xray

Similarity Score: 0.2338

Result 4

Image ID: ImageCLEFmedCaption_2022_train_000939

Caption: upper gastrointestinal contrast xray check of esophagojejunostomy at postoperative

Similarity Score: 0.2215

Result 5

Image ID: ImageCLEFmedCaption_2022_train_000516

Caption: thoracic spine anteroposterior xray show the retain epidural catheter arrow

Similarity Score: 0.2168

Medical Image Caption Retrieval System

Top 5 Results for: "CT scan demonstrating pulmonary embolism"

Result 1

Image ID: ImageCLEFmedCaption_2022_train_000892

Caption: dripinfusion cholangiographycompute tomography scan fortyeight day postoperatively the right accessory hepatic duct be intact black arrow and it dominate an anterior segment of the right hepatic lobe as indicate by the intraoperative cholangiography scan

Similarity Score: 0.3283

Result 2

Image ID: ImageCLEFmedCaption_2022_train_000840

Caption: computed tomography scan show the presence of a right heart tumor develop on both side of the tricuspid valve

Similarity Score: 0.248

Result 3

Image ID: ImageCLEFmedCaption_2022_train_000211

Caption: man with severe aortic stenosis doubleoblique reconstruction image of ecggate multidetector ct show calcified tricuspid valve

Similarity Score: 0.2241

Result 4

Image ID: ImageCLEFmedCaption_2022_train_000930

Caption: a conventional coronary angiographic image at a right anterior obliquecaudal view two aneurysm be homogenously enhance by contrast a and b the large aneurysm show irregular enhancement in the periphery black arrow and most of the central bea be fill with thrombus c the second aneurysm of the left circumflex artery show homogenous filling with contrast material b faint contrast stain be note on the dbetal side of the large aneurysm white arrow a delay image might have show collateral that be see on ct angiography ra ramus intermedius ct compute tomography

Similarity Score: 0.2075

Result 5

Image ID: ImageCLEFmedCaption_2022_train_000994

Caption: preoperative aortogram with occlusion of the superior mesenteric artery and high grade stenosis of celiac and inferior mesenteric artery arrow note the markedly hypertrophied ima which collateralise with the sma via the marginal branch of drummond arrowhead and the arc of riolan

Similarity Score: 0.0